

Strategic Content Creation with GenAI: To Share or Not to Share?

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Abstract

We introduce a game-theoretic framework that examines strategic interactions between a platform and its content creators in the presence of AI-generated content. Our model’s main novelty is in capturing creators’ dual strategic decisions: their investment in content quality and their (possible) consent to share their content with the platform’s GenAI, both of which significantly impact their utility. To incentivize creators, the platform strategically allocates a portion of its GenAI-driven revenue to creators who share their content. We focus on the class of *full-sharing* equilibrium profiles, in which all creators willingly share their content with the platform’s GenAI system. Such equilibria are highly desirable both theoretically and practically. Our main technical contribution is formulating and efficiently solving a novel optimization problem that approximates the platform’s optimal revenue subject to inducing a full-sharing equilibrium. A key aspect of our approach is identifying conditions under which full-sharing equilibria exist and a surprising connection to the Prisoner’s Dilemma. Finally, our simulations demonstrate how revenue-allocation mechanisms affect creator utility and the platform’s revenue.

CCS Concepts

• **Theory of computation** → **Algorithmic game theory**; *Algorithmic mechanism design*; • **Applied computing** → *Economics*.

Keywords

game theory, content platform, Stackelberg game, mechanism design, strategic behavior, multi-agent systems, GenAI, equilibrium

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1 Introduction

Online content platforms such as YouTube, Medium, and SoundCloud have become ubiquitous, hosting vast ecosystems of creators who aim to maximize their reach, engagement, and revenue. The decision-making challenges faced by these creators have been the focus of a growing body of work, examining aspects such as revenue-sharing mechanisms [9], fairness in content visibility and

diversification [8, 32], and the economic incentives shaping content creation [2, 6, 7, 15, 26]. As these platforms evolve, especially with advances in Generative AI (GenAI), new considerations arise.

GenAI enables platforms to synthesize content by distilling and recombining existing creator contributions. For instance, a news platform could use GenAI tools to generate personalized summaries of the day’s news tailored to individual users. This enhances user satisfaction and opens new monetization channels. However, it also introduces two significant challenges: sharing consent and content quality.¹ First, using creator content for GenAI raises legal and ethical questions around intellectual property and copyright infringement [36]. Second, AI-generated content may cannibalize attention from human-created work, reducing creators’ incentives to invest in content quality.

To address these challenges, platforms may offer a remedy: sharing GenAI-driven revenue with creators who allow the platform’s GenAI to use their content. This mirrors existing revenue-sharing programs on platforms like YouTube and X (formerly Twitter) [42, 53], and could incentivize both sharing consent and content quality. The main question thus becomes: *how should the platform allocate its GenAI-driven revenue to maximize its (retained) revenue while incentivizing the creation and sharing of high-quality content?*

This paper pioneers the study of revenue sharing in the presence of strategic content creators and AI-generated content. We present a game-theoretic model consisting of a platform and strategic creators, in which the platform can generate content using GenAI tools and allocate its GenAI-driven revenue based on creator actions. Our framework incorporates creators’ dual strategic decisions: (i) the extent to which they share their content with the platform’s GenAI, ranging from full withholding to full sharing, and (ii) the body of content they create, each incurring its own cost. The platform selects a revenue-sharing mechanism rewarding creators based on these two axes of contribution, with the goal of optimizing for content quality, overall revenue, or a tradeoff between them.

1.1 Our Contribution

Our contribution is twofold. First and foremost, we introduce the first game-theoretic model that captures both content production and data sharing in Generative AI ecosystems, where creators *and* the platform are strategic. User traffic is allocated through a Tullock contest structure [40], where creators and the platform’s GenAI system compete based on their relative content quality. Within this environment, each creator decides not only the quality of their content, but also how much of it to share with the GenAI. The GenAI’s content quality is determined by the pool of shared creator content; thus, the platform should incentivize creators to produce and share content. To that end, we assume the platform can select

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¹For instance, YouTube allows its creators to opt out of having their content used for GenAI training [52].

a proportional allocation rule (see Equation (1)) parametrized by $\rho \in [0, 1]$, and redistribute a ρ -portion of its GenAI-driven revenue to the creators. The platform is also strategic, and can choose ρ in a way that steers creators towards revenue-maximizing outcomes.

Our second contribution is technical. We study equilibria in the induced game, and focus on a subclass of equilibrium profiles we term *full-sharing equilibria* (FSE for brevity; see Definition 1). In an FSE, creators are better off by sharing their content. FSEs are desirable due to several properties that we discuss in Section 2.1. We present a comprehensive characterization of FSE in Section 3, showing that any ρ induces at most one FSE. In Section 4, we study the platform's revenue optimization problem (see PO). The platform can choose any ρ to maximize its revenue under the induced FSE, if it exists. Such bilevel optimization is notoriously NP-hard [25], and in our case, the difficulty is further exacerbated. Among other factors, finding equilibrium is non-trivial as best-response computation is non-concave, and the domain of feasible solutions does not have a closed-form. Nevertheless, we show the following:

THEOREM 1 (INFORMAL VERSION OF THEOREM 8). *There exists an efficient algorithm that, for any $\varepsilon > 0$, computes a revenue allocation inducing an ε -FSE and achieving platform revenue within ε of the maximum attainable under any FSE.*

Our approach hinges on smoothness analysis of fixed points in an auxiliary game we call *enforced sharing game*, where creators are forced to share their content. Notably, this result is not limited to revenue maximization; we show in Section 6 that our algorithm extends to other platform objectives. Finally, we present synthetic experiments to examine how revenue-allocation mechanisms affect content production, creator utility, and the platform's revenue.

Due to space constraints, we defer all formal proofs, additional extensions, and supplementary simulation results to the full version of this paper [31].

1.2 Related Work

Strategic content provider dynamics have received substantial attention in recent years, with numerous models analyzing how creators compete for attention and adapt their strategies in algorithmically mediated ecosystems [2, 6, 7, 15, 26, 28]. These works address key challenges such as incentivizing content quality [23, 27, 47, 49], ensuring fairness and diversity [3, 16, 32, 45, 46], and aligning creator incentives with platform-level objectives [11, 54]. The recent proliferation of Generative AI (GenAI) has introduced new complexity into these ecosystems, prompting a line of research exploring how GenAI alters competitive dynamics. For instance, some works consider environments in which either GenAI or human agents act strategically, but not both [18, 38], while others examine GenAI systems that act strategically to preserve human participation [39].

The work most closely related to ours is Yao et al. [48], who introduce a model of competition between human creators and a GenAI agent within a Tullock contest [14, 19, 20, 22], a well-established economic model of competition. Our model, like that of Yao et al. [48], captures how user traffic is distributed based on the relative quality of content produced by humans and a GenAI system. The effectiveness of the GenAI system itself is derived from human-generated content. However, our paper significantly departs from Yao et al. [48]. First, we allow creators to decide whether and to

what extent their content is shared with the GenAI system, making data sharing itself a strategic choice. Second, we model the platform as a strategic actor that designs a revenue-allocation mechanism to influence creator behavior. These modifications result in a doubly strategic setting, where both creators and the platform engage in interdependent optimization. From a technical perspective, the creators' strategy spaces become more complex, and the game is no longer monotone [34]. Furthermore, modeling the platform as a strategic entity gives rise to a novel bilevel optimization problem of revenue maximization, which has not been addressed in previous works. Consequently, our research questions are novel, and our results significantly extend the existing literature by addressing strategic interactions that Yao et al. [48] have not explored.

More broadly, since our work addresses (strategic) content sharing with GenAI, it relates to works on data valuation [4, 29, 41] and data markets [1]. Furthermore, we consider the allocation of GenAI-driven revenue to content creators, thereby relating to copyright challenges with GenAI [21, 33, 44], attribution to specific data sources [10], and royalties [17].

2 Model

We consider a platform that hosts content produced by a set of human creators, alongside content generated by a GenAI system controlled by the platform. We formally define the components of the interaction below.

Creators. We consider $n \geq 2$ human content creators competing for user traffic and denote the set of creators by $[n] = \{1, 2, \dots, n\}$, with a typical creator indexed by $i \in [n]$. Each creator produces a *body of content* that encompasses multiple attributes: quality, quantity, trend alignment, domain expertise or specialty, and so on. As is standard in prior work (e.g., [48, 50]), we aggregate these attributes into a single scalar, and without loss of generality, call it *content quality*. Formally, creator i selects a quality $x_i \in [0, \infty)$.

For each creator i , creating content x_i incurs a production cost $c_i(x_i)$. While we do not assume a particular form of the cost function, we have several structural assumptions that are standard in prior work. We assume that for every $i \in [n]$, the cost function c_i is non-decreasing, twice differentiable, and strictly convex. Moreover, we assume that $c_i(0) = 0$, $\lim_{x \rightarrow 0^+} dc_i(x)/dx = 0$, and $\lim_{x \rightarrow \infty} dc_i(x)/dx = \infty$. These assumptions ensure that not creating content is costless, and that the marginal cost of improving the quality by one unit is initially zero but becomes unbounded.

Beyond choosing the content quality, each creator also chooses their *sharing level* $s_i \in [0, 1]$. The sharing level represents the extent to which creator i allows their content x_i to be used by the GenAI system, ranging from complete withholding ($s_i = 0$) to full sharing ($s_i = 1$). Therefore, the strategy of each creator $i \in [n]$ is a pair (x_i, s_i) . Let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{s} = (s_1, \dots, s_n)$ denote the quality and sharing profiles, respectively. Additionally, for a creator $i \in [n]$, we let $\mathbf{x}_{-i} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ and $\mathbf{s}_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$ denote the quality and sharing profiles after omitting the i 'th entry.

Generative AI. The platform operates a GenAI system, modeled as a non-strategic agent whose output quality depends on the aggregate quality of the content shared from creators, $\mathbf{x}^\top \mathbf{s} = \sum_{j=1}^n x_j s_j$.

Its quality is given by $Q_{AI}(\mathbf{x}, \mathbf{s}) = \alpha \cdot \mathbf{x}^\top \mathbf{s}$, where $\alpha > 0$ represents the efficiency of data usage.² Notably, if no data is shared (i.e., $\mathbf{s} = 0$), then $Q_{AI} = 0$.

User traffic and engagement. The platform attracts user traffic denoted by $T(\mathbf{x})$, which increases monotonically with the total quality of human content, $\|\mathbf{x}\| = \sum_{j=1}^n x_j$, where we use $\|\cdot\|$ without a subscript to denote the ℓ_1 norm throughout the paper. We model traffic as $T(\mathbf{x}) = \mu \cdot \|\mathbf{x}\|^\gamma$, where $\mu > 0$ is a baseline traffic parameter and $\gamma \in [0, 1]$ captures the elasticity of traffic with respect to the total human content quality [13, 37].³

Given a strategy profile $(\mathbf{x}, \mathbf{s})$ and the induced user traffic $T(\mathbf{x})$, users allocate their attention to human and AI-generated content according to a Tullock-like decision based on relative quality [40]. Specifically, the fraction of attention allocated to creator i 's content is $\frac{x_i}{\|\mathbf{x}\| + Q_{AI}(\mathbf{x}, \mathbf{s})}$, while the fraction allocated to AI-generated content is $\frac{Q_{AI}(\mathbf{x}, \mathbf{s})}{\|\mathbf{x}\| + Q_{AI}(\mathbf{x}, \mathbf{s})}$. In both cases, the denominator represents the total quality of created and generated content available on the platform.

Platform strategy. The platform may have multiple sources of revenue. However, to ease presentation, we focus on the user traffic it receives from its AI-generated content. Note that it depends on Q_{AI} , which in turn depends on human content. To obtain high-quality AI-generated content, the platform can allocate some of this revenue back to creators. Formally, the platform commits to an *allocation rule* $f : [0, \infty)^n \times [0, 1]^n \rightarrow \Delta^{n+1}$. It maps every strategy profile $(\mathbf{x}, \mathbf{s})$ to a vector $(f_1(\mathbf{x}, \mathbf{s}), \dots, f_n(\mathbf{x}, \mathbf{s}), f_P(\mathbf{x}, \mathbf{s}))$, where $f_i(\mathbf{x}, \mathbf{s})$ represents the fraction of GenAI-driven revenue allocated to creator $i \in [n]$, and $f_P(\mathbf{x}, \mathbf{s})$ is the fraction retained by the platform. Here, Δ^{n+1} denotes the $(n+1)$ -dimensional simplex.

While various allocation mechanisms are possible, we focus on the class of parameterized *proportional allocation rules* f_ρ , formally defined below, and address other designs in Section B. The revenue-allocation parameter $\rho \in [0, 1]$, or simply the *allocation parameter*, specifies the fraction of GenAI-driven revenue allocated to creators, with the remainder $1 - \rho$ retained by the platform. Under this rule, the platform distributes the ρ -fraction among creators in proportion to their effective shared contributions $x_i s_i$, yielding

$$f_{i,\rho}(\mathbf{x}, \mathbf{s}) = \begin{cases} \rho \cdot \frac{x_i s_i}{\mathbf{x}^\top \mathbf{s}} & \mathbf{x}^\top \mathbf{s} > 0, \\ 0 & \mathbf{x}^\top \mathbf{s} = 0 \end{cases} \quad (1)$$

and $f_{P,\rho}(\mathbf{x}, \mathbf{s}) = 1 - \sum_{i=1}^n f_{i,\rho}(\mathbf{x}, \mathbf{s})$. We note that whenever $\mathbf{x}^\top \mathbf{s} > 0$, it holds that $\sum_{i=1}^n f_{i,\rho}(\mathbf{x}, \mathbf{s}) = \rho$ and $f_{P,\rho}(\mathbf{x}, \mathbf{s}) = 1 - \rho$.

Revenue and utilities. Each human creator $i \in [n]$ receives revenue from two sources and incurs a creation cost. The first revenue source is direct user traffic to their content, yielding a revenue of $T(\mathbf{x}) \frac{x_i}{\|\mathbf{x}\| + Q_{AI}}$. The second revenue source is their allocated share of the GenAI-driven revenue, given by $T(\mathbf{x}) \frac{Q_{AI}}{\|\mathbf{x}\| + Q_{AI}} \cdot f_{i,\rho}(\mathbf{x}, \mathbf{s})$. Overall, creator i 's utility is

$$U_i(\mathbf{x}, \mathbf{s}; f_\rho) = T(\mathbf{x}) \frac{x_i}{\|\mathbf{x}\| + Q_{AI}} + T(\mathbf{x}) \frac{Q_{AI}}{\|\mathbf{x}\| + Q_{AI}} \cdot f_{i,\rho}(\mathbf{x}, \mathbf{s}) - c_i(x_i).$$

²We discuss extensions of some of our results to a more general functional form of Q_{AI} in Section B.

³We assume that users are primarily attracted to human-generated content [5]. We relax this assumption and analyze GenAI-dependent traffic in Section B.

The platform's revenue corresponds to the portion of GenAI revenue it retains under the allocation rule f_ρ , which is given by

$$U_P(\mathbf{x}, \mathbf{s}; f_\rho) = T(\mathbf{x}) \frac{Q_{AI}}{\|\mathbf{x}\| + Q_{AI}} \cdot f_{P,\rho}(\mathbf{x}, \mathbf{s}).$$

Stackelberg game. We model the interaction as a Stackelberg game, with the platform as the leader and the creators as followers. The platform first commits to a revenue-allocation rule f_ρ . This choice induces a subgame among the creators, denoted by $G(\rho)$. In this game, each creator $i \in [n]$ simultaneously selects a strategy (x_i, s_i) to maximize their utility given the fixed rule f_ρ . The resulting profile $(\mathbf{x}, \mathbf{s})$ determines the user traffic $T(\mathbf{x})$, the quality of AI-generated content $Q_{AI}(\mathbf{x}, \mathbf{s})$, and the resulting payoffs $U_i(\mathbf{x}, \mathbf{s}; f_\rho)$ and $U_P(\mathbf{x}, \mathbf{s}; f_\rho)$. This structure enables the platform to steer creator behavior through the choice of the allocation parameter ρ .

2.1 Solution Concepts

We are interested in stable outcomes in the subgame $G(\cdot)$, where stability refers to pure Nash equilibrium (PNE). We highlight a class of PNEs that we call *Full Sharing PNE*, or FSE for brevity. In FSE, full content sharing arises endogenously: each creator chooses to share their content because the platform's revenue-allocation mechanism renders sharing a best response.

Definition 1 (FSE). A strategy profile $(\mathbf{x}, \mathbf{1})$ is an FSE in the subgame $G(\rho)$ if for every creator $i \in [n]$,

$$U_i(\mathbf{x}, \mathbf{1}; f_\rho) \geq \sup_{x'_i \in [0, \infty), s'_i \in [0, 1]} U_i((x'_i, \mathbf{x}_{-i}), (s'_i, \mathbf{1}_{-i}); f_\rho).$$

FSEs are a special case of PNEs in which all creators fully share their content. While other equilibria involving partial or no sharing may exist, FSEs are appealing from the platform's perspective: they maximize the quality of AI-generated content given the created content. Focusing on FSE is also practically motivated, as it prevents unauthorized use of content and disputes or resentment from creators who believe their content was used without consent. FSEs naturally extend to approximate equilibria: a strategy profile $(\mathbf{x}, \mathbf{1})$ is an ε -FSE if no creator i has an ε -beneficial deviation.

To facilitate the analysis in the forthcoming sections, we also present the following auxiliary equilibrium concept called *Enforced Sharing PNE*, or simply ESE.

Definition 2 (ESE). A quality profile $\mathbf{x}$ is an ESE in the subgame $G(\rho)$ if for every creator $i \in [n]$,

$$U_i(\mathbf{x}, \mathbf{1}; f_\rho) \geq \sup_{x'_i \in [0, \infty)} U_i((x'_i, \mathbf{x}_{-i}), \mathbf{1}; f_\rho).$$

To motivate Definition 2, imagine the auxiliary *enforced sharing game*, where f_ρ is fixed and creators *must* share their content, i.e., $\mathbf{s} = \mathbf{1}$. ESEs allow us to analyze game dynamics by focusing solely on creators' quality profiles $\mathbf{x}$. We remark that any quality profile $\mathbf{x}$ such that $(\mathbf{x}, \mathbf{1})$ is an FSE also constitutes an ESE.

3 FSE Analysis

In this section, we analyze how creators respond to an allocation rule. Subsection 3.1 assumes a fixed allocation rule f_ρ and studies the structure of ESEs in the subgame $G(\rho)$. We show that every ρ induces a unique ESE, although different values of ρ may induce different ESEs. In Subsection 3.2, we move to study FSEs. Namely,

Theorem 5 characterizes a sufficient condition under which the unique ESE $\mathbf{x}^*$ also constitutes the unique FSE $(\mathbf{x}^*, 1)$.

3.1 Optimal Sharing and Unique ESE

We begin by analyzing the creators' best responses under a fixed allocation rule f_ρ . First, we provide a full characterization of the optimal sharing level s_i for a given profile $(\mathbf{x}, \mathbf{s}_{-i})$, showing that it takes a threshold form.

LEMMA 2. *Fix an arbitrary creator i with content $x_i > 0$, and a profile $(\mathbf{x}_{-i}, \mathbf{s}_{-i})$. Further, let τ_i such that $\tau_i = \frac{x_i}{x_i + \|\mathbf{x}_{-i}\| + \alpha \mathbf{x}_{-i}^\top \mathbf{s}_{-i}}$. The optimal sharing level s of creator i w.r.t. $(x_i, \mathbf{x}_{-i}, \mathbf{s}_{-i})$, i.e., $s \in \arg \max_{s \in [0,1]} U_i((x_i, \mathbf{x}_{-i}), (s, \mathbf{s}_{-i}); f_\rho)$ is given by*

$$s^* = \begin{cases} 1 & \rho > \tau_i \\ \text{any number from } [0, 1] & \rho = \tau_i \\ 0 & \rho < \tau_i \end{cases} \quad (2)$$

The proof of Lemma 2 involves showing that creator i 's utility is strictly increasing in s_i when $\rho > \tau_i$, strictly decreasing when $\rho < \tau_i$, and constant when $\rho = \tau_i$, yielding Equation (2).

Lemma 2 reveals an interesting asymmetry: creators producing a large share of $\|\mathbf{x}\|$ have higher thresholds; thus, they are more inclined to withhold their data, as doing so weakens a powerful GenAI competitor. In contrast, creators with relatively low quality have stronger incentives to share. Notably, the sharing decision becomes effectively binary—each creator either fully shares or fully withholds, with no incentive to choose an intermediate level.

Another insight from Lemma 2 concerns the case $\mathbf{s} = \mathbf{0}$, where all creators initially withhold their data. Fix a quality profile $\mathbf{x}$, and notice that if the number of creators n is moderately large, the threshold $\tau_i = \frac{x_i}{\|\mathbf{x}\|}$ becomes small for most creators. Consequently, even a low revenue-allocation parameter $\rho \approx 1/n$ can make it beneficial for the least productive creator, denoted by i , to deviate and set $s_i = 1$. This deviation increases $\mathbf{x}^\top \mathbf{s}$, thereby lowering the threshold τ_j for other creators $j \neq i$ to $\frac{x_j}{\|\mathbf{x}\| + \alpha x_i} < \frac{x_j}{\|\mathbf{x}\|} = \tau_j$. As a result, creator i 's deviation can potentially trigger a cascade of full-sharing responses from other players, an effect similar to the classic prisoner's dilemma.

Next, we study creators' quality choices under enforced sharing.

LEMMA 3. *For any fixed ρ , $G(\rho)$ admits a unique ESE.*

The proof relies on Rosen's diagonal strict concavity (DSC) condition [34], a standard approach for establishing existence and uniqueness of PNEs in concave games.

Given the existence and uniqueness established in Lemma 3, we can define the ESE correspondence: let $\mathcal{X} : [0, 1] \rightarrow [0, \infty)^n$ denote the mapping from ρ to its corresponding unique ESE profile, $\mathcal{X}(\rho)$, in the subgame $G(\rho)$.

3.2 Existence of FSE

We now examine conditions under which full sharing can arise in a PNE. For a profile to be an FSE, it must induce a quality profile that constitutes an ESE under enforced sharing. The uniqueness of the ESE implies that, if an FSE exists, both the creators' content qualities and their sharing decisions are uniquely determined. Therefore, the

remaining question is whether the provided revenue allocation is enough to induce such a profile in equilibrium.

First, we consider the two extreme cases, $\rho = 0$ and $\rho = 1$. The former is trivial: creators can only be worse off from sharing; hence, withholding content is a dominant strategy, and no FSE exists in $G(0)$. The latter case is more interesting. If $\rho = 1$, the platform allocates its entire GenAI-driven revenue to the creators. While this might not be practical, it provides a useful reference point for creator-centered ecosystems.

PROPOSITION 4. *$G(1)$ admits a unique PNE, which is an FSE.*

The result builds on a key structural property: when $\rho = 1$, full sharing becomes a dominant strategy for every creator. Lemma 2 offers intuition: as long as $\mathbf{x}$ is strictly positive, $\rho = 1$ will surpass every possible threshold τ_i , no matter the choice of x_i . This guarantees that any PNE must involve full sharing, and existence and uniqueness of the resulting quality profile then follow from Lemma 3. Extensions of Proposition 4 to other forms of Q_{AI} are discussed in Section 6.

Generalizing Proposition 4 beyond $\rho = 1$ requires careful attention. Although Lemmas 2 and 3 provide a tight characterization of the (decoupled) best responses in s and $\mathbf{x}$, the main challenge is ensuring that no combined deviation in both is beneficial. The following Theorem 5 offers a sufficient condition for a value of ρ to induce an FSE.

THEOREM 5. *Fix any $\rho \in [0, 1]$. For every $i \in [n]$, define $x_i^{\max}$ to be the unique positive value satisfying the equation $c_i(z) = \mu z^\gamma$. If $\rho > \max_{i \in [n]} \left\{ \frac{x_i^{\max}}{x_i^{\max} + (1+\alpha)\|\mathcal{X}_{-i}(\rho)\|} \right\}$, then $(\mathcal{X}(\rho), 1)$ is the unique FSE of the subgame $G(\rho)$.*

PROOF SKETCH OF THEOREM 5. Fix some creator i . The proof relies on $x_i^{\max}$, a quality such that any choice of $x_i > x_i^{\max}$ will result in a strictly negative utility for i , regardless of its sharing level. When $\mathbf{s}_{-i} = 1_{-i}$, the threshold from Lemma 2 becomes $\frac{x_i}{x_i + (1+\alpha)\|\mathbf{x}_{-i}\|}$. This threshold is strictly increasing in x_i ; thus, if $\rho > \frac{x_i^{\max}}{x_i^{\max} + (1+\alpha)\|\mathcal{X}_{-i}(\rho)\|}$, any profitable deviation of i must involve $s_i = 1$. If this holds for every i , Lemma 3 guarantees no profitable deviations in $\mathbf{x}$ exist as well. $\square$

3.3 Analysis under Power Cost Functions

Although the condition in Theorem 5 is theoretically meaningful, it is not constructive. The key challenge is that the quantity $\|\mathcal{X}_{-i}(\rho)\|$, determining the right-hand side of the inequality, depends on the ESE $\mathcal{X}(\rho)$, which in turn is a function of ρ . Generally, it is not possible to obtain a closed-form expression for $\mathcal{X}(\rho)$ in terms of ρ , making the inequality difficult to evaluate. To enable more precise analysis, we focus on a tractable yet flexible class of *power cost functions*, a standard structure in economic and game-theoretic models of effort and production [35, 48, 51].

Definition 3. An instance is called $(n, a_{\min}, a_{\max}, \theta)$ -power cost if it consists of n creators, where each creator $i \in [n]$ has a cost function $c_i(x_i) = a_i x_i^\theta$ with common exponent $\theta \in (1, 2]$ and cost coefficients $a_i \in [a_{\min}, a_{\max}]$.

Power cost functions enable more precise analysis. The next proposition reveals that as creator ecosystems grow large, inducing

full sharing becomes increasingly tractable, requiring only modest revenue-sharing commitments from the platform.

PROPOSITION 6. *For any $a_{\min}, a_{\max}, \theta$ and $w \in (0, 1]$, there exists $N(w) \in \mathbb{N}$ such that for all $n > N(w)$, in any $(n, a_{\min}, a_{\max}, \theta)$ -power cost instance, the subgame $G(\rho)$ admits an FSE for every $\rho \in (w, 1]$.*

The proof of Proposition 6 involves the construction of a lower bound on $\|\mathcal{X}_{-i}(\rho)\|$ that grows with the number of creators n . We then obtain the desired result by substituting this lower bound in Theorem 5 and extracting the necessary n .

While Proposition 6 suggests that platforms can achieve FSE with low revenue sharing, this raises a natural question about content production incentives. Lower allocation parameters may reduce creators' motivation to invest in quality, potentially decreasing overall platform traffic. The next proposition examines this relationship.

PROPOSITION 7. *Fix any homogeneous power cost instance, i.e., (n, a, a, θ) for some $a > 0$. Then, $\|\mathcal{X}(\rho)\|$ is strictly increasing in ρ .*

Proposition 6 and Proposition 7 reveal competing incentives in the choice of ρ : retaining a larger share of existing GenAI revenue versus increasing total revenue through larger traffic. The next section navigates this tradeoff algorithmically.

4 Platform Revenue Maximization

We now turn to the platform's strategic problem of selecting the revenue-allocation parameter $\rho \in [0, 1]$ that maximizes its retained revenue, subject to the constraint that creators respond by fully sharing their data. This induces a bilevel optimization problem: the platform acts as a Stackelberg leader, anticipating the equilibrium behavior of creators in response to its choice of ρ .

4.1 Formulation and Relaxation

The platform's optimization problem, which we term **Platform Optimization**, is given by:

$$\sup_{\rho \in [0,1]} U_P(\mathcal{X}(\rho), 1; f_\rho) \quad \text{subject to } (\mathcal{X}(\rho), 1) \text{ is an FSE.} \quad (\text{PO})$$

Further, let OPT denote the supremum, which is not necessarily attainable by any feasible solution. Solving PO is challenging for two main reasons. First, checking whether a specific ρ induces an FSE requires solving a complex fixed-point problem involving a generally non-convex system of equations, making exact computation of FSE intractable in most cases. Second, even with the ability to verify the above efficiently, optimizing $U_P(\mathcal{X}(\rho), 1; f_\rho)$ over ρ analytically is nontrivial due to its implicit dependence on $\mathcal{X}(\rho)$. Practical approaches might involve a grid search over ρ , but this approach does not guarantee the approximate solution will be feasible with respect to the FSE constraint. To that end, we relax the constraint and focus on approximate FSEs. Finally, to allow efficient approximation, we impose additional regularity assumptions on the cost functions by assuming they are β -strongly convex—a standard condition in strategic optimization [24, 43].

4.2 Solution Methods

First, we propose a simple search-based procedure for computing an approximate solution of PO, illustrated in Algorithm 1. It receives

Algorithm 1 Approximation Algorithm for PO

Input: Game instance $\mathcal{I}, \delta, \eta$

- 1: Initialize $\hat{\rho} \leftarrow 0$ and $\hat{U} \leftarrow 0$
- 2: **for** $t = 1 \dots \lceil 1/\delta \rceil$ **do**
- 3: Let $\rho \leftarrow \min \left\{ \frac{t}{\lceil 1/\delta \rceil}, 1 \right\}$
- 4: Find $\mathbf{x}$ such that $\|\mathbf{x} - \mathcal{X}(\rho)\|_2 \leq \delta$
- 5: **if** $(\mathbf{x}, 1)$ is an η -FSE in $G(\rho)$ **then**
- 6: **if** $U_P(\mathbf{x}, 1; f_\rho) > \hat{U}$ **then**
- 7: $\hat{\rho} \leftarrow \rho$ and $\hat{U} \leftarrow U_P(\mathbf{x}, 1; f_\rho)$
- 8: **return** $\hat{\rho}$

as input an instance $\mathcal{I}$ and two parameters, δ and η . We carefully determine the values of δ and η to reach a certain approximation level; the exact values are given in Theorem 8. Algorithm 1 performs a grid search over the interval $[0, 1]$, evaluating a discrete set of allocations parameters at resolution δ . Starting with initialized variables (Line 1), the algorithm iterates through candidate values $\rho = t/\lceil 1/\delta \rceil$ (Lines 2–7). For each candidate, it approximates the corresponding ESE (Line 4) and verifies whether it constitutes an η -FSE (Line 5). If this stability criterion is met, it updates the current best solution whenever the resulting platform revenue exceeds the previous maximum (Line 7). The algorithm ultimately returns the best-performing ρ across all valid candidates (Line 8). The next result establishes that the algorithm returns a near-optimal solution with provable guarantees on both approximation quality and runtime.

THEOREM 8. *Fix any instance $\mathcal{I}$ with strongly convex cost functions. There exist constants A, B such that for any $\varepsilon > 0$, the execution of Algorithm 1 with $\delta = \frac{\varepsilon}{A}$ and $\eta = \frac{\varepsilon}{4}$ can be implemented in time $O\left(\frac{A^2 B}{\varepsilon^2}\right)$. Moreover, its output $\hat{\rho}$ induces an ε -FSE and satisfies $U_P(\mathcal{X}(\hat{\rho}), 1; f_{\hat{\rho}}) \geq \text{OPT} - \varepsilon$.*

The exact specification of $A = A(\mathcal{I})$ and $B = B(\mathcal{I})$ involves somewhat technical details; however, both constants depend on upper and lower bounds of the equilibrium quality $\|\mathcal{X}(\rho)\|$, which under mild regularity assumptions are themselves polynomial in the number of creators n . For the special case of power costs (recall Definition 3), we obtain:

PROPOSITION 9. *Fix any $(n, a_{\min}, a_{\max}, \theta)$ -power cost instance $\mathcal{I}$ with $a_{\min}, a_{\max} = O(1)$. Then the constants $A(\mathcal{I})$ and $B(\mathcal{I})$ from Theorem 8 satisfy $A(\mathcal{I}) = O(n^2)$ and $B(\mathcal{I}) = O(n)$.*

Although Algorithm 1 offers an efficient method for achieving an anytime approximation of OPT, its runtime inevitably depends on n . Thus, for large platforms, it may be preferable to use a fixed allocation rule rather than find one algorithmically, as the latter option might be impractical. It turns out that for many power cost instances, there is a closed-form choice of ρ that induces an FSE and provides a multiplicative approximation of the platform's optimal revenue. Interestingly, this choice of ρ does not depend on n or the coefficients $(a_i)_{i=1}^n$.

PROPOSITION 10. *Consider large enough n, μ and any power cost instance such that $(\alpha + 1)\gamma > \theta$. Then, choosing $\rho = (\alpha\gamma + \gamma - \theta)/\alpha\theta$*

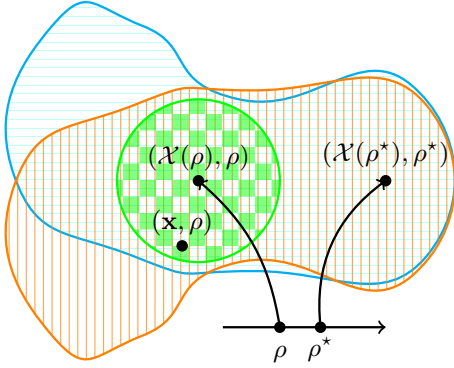


Figure 1: Illustrating the proof sketch.

ensures that ρ induces an FSE, and yields platform revenue satisfying $U_\rho(X(\rho), 1; f_\rho) \geq (2\alpha + 2)^{-\frac{\gamma}{\theta-\gamma}} \cdot \text{OPT}$.

Notably, larger values of θ decrease the expression of ρ in Proposition 10, and, perhaps counterintuitively, larger values of α increase it. We detail more about these phenomena in Section 5, where they also arise from simulations on the optimal allocation parameter. The proof of Proposition 10 builds on a lower bound of $\|X(\rho)\|$ from [48], which we adapt to our setting. We then derive the closed-form expression for ρ by substituting this lower bound into the platform's revenue function and optimizing it analytically. Proposition 6 ensures that the resulting profile is a valid FSE, while the approximation factor reflects the gap between the bound and the true value of $\|X(\rho)\|$.

4.3 Proof Sketch of Theorem 8

Fix any instance $\mathcal{I}$ and $\varepsilon > 0$, and let δ, η denote the parameters in Theorem 8 that are passed to the algorithm. Next, focus on any feasible solution ρ^* to PO, and let ρ denote the closest value to ρ^* the algorithm considers in Line 3. Notably, $|\rho - \rho^*| \leq \delta$. Figure 1 illustrates how ρ^* is mapped to the unique ESE of $G(\rho^*)$, $X(\rho^*)$. Similarly, ρ is mapped to $X(\rho)$, the unique ESE of $G(\rho)$.

Step 1: Beneficial deviations. We define the function ϕ such that $\phi(\tilde{x}, \tilde{\rho})$ is the maximal gain from any creator's deviation from $(\tilde{x}, 1)$ in $G(\tilde{\rho})$. Because ρ^* is feasible, $(X(\rho^*), 1)$ is the FSE of $G(\rho^*)$; therefore, $\phi(X(\rho^*), \rho^*) = 0$. Using technical lemmas, we show that ϕ is Lipschitz continuous and bound its Lipschitz constant. Additionally, we use the Implicit Function Theorem to bound the Lipschitz constant of $X(\cdot)$, circumventing its lack of a closed-form expression. The blue region in Figure 1 (filled with horizontal lines) depicts the set of pairs $(\tilde{x}, \tilde{\rho})$ such that $\phi(\tilde{x}, \tilde{\rho}) \leq \eta < \varepsilon$. A careful selection of δ implies that ρ and ρ^* are close, and so are $X(\rho)$ and $X(\rho^*)$; therefore, $(X(\rho), \rho)$ lies within the blue region, implying that the condition in Line 5 holds and ρ is an ε -FSE of $G(\rho)$.

Step 2: Revenue approximation. Similarly, we show that the function $U_\rho(\tilde{x}, 1; f_\rho)$ is Lipschitz continuous and upper bound its Lipschitz constant. This ensures that there exists a neighborhood around $(X(\rho^*), \rho^*)$ with pairs $(\tilde{x}, \tilde{\rho})$ that obtain ε -close revenue. We let the orange region in Figure 1, filled with vertical lines, depict all

such pairs. Picking δ appropriately ensures that $(X(\rho), \rho)$ belongs to the orange region. Thus, in case ρ^* achieves a revenue arbitrarily close to OPT, at least one nearby point ρ on the discrete grid yields a revenue within ε of optimal.

Step 3: Resolving output mismatch. Although we have demonstrated in Steps 1 and 2 that $X(\rho)$ satisfies both the ε -FSE and revenue guarantees when ρ lies sufficiently close to ρ^* , the algorithm does not evaluate $X(\rho)$ directly. Instead, it computes an approximate profile $\mathbf{x}$ such that $\|\mathbf{x} - X(\rho)\|_2 \leq \delta$ (recall Line 4). This creates a mismatch: even if $X(\rho)$ is a valid and near-optimal solution, the algorithm might overlook it because the nearby profile $\mathbf{x}$ fails one of the checks in Lines 5 and 6. To resolve this, let the green region (with a checkerboard pattern) in Figure 1 denote all pairs $(\tilde{x}, \rho)$ such that $\|\tilde{x} - X(\rho)\|_2 \leq \delta$. Unlike the orange and blue regions, the green region is a Euclidean ball. Note that $(\mathbf{x}, \rho)$ is in this region for any arbitrary $\mathbf{x}$ that may have been found in Line 4. By choosing δ small enough so that the green ball is fully contained within both the blue and orange regions, we ensure that all such $\mathbf{x}$ exhibit the same stability and revenue guarantees as $X(\rho)$, and vice versa. Thus, the algorithm safely evaluates $\mathbf{x}$ in place of $X(\rho)$, and its output ρ satisfies the desired approximation guarantees.

Efficient computation. First, we prove that the enforced sharing game belongs to the class of strongly monotone games [34]. This property allows us to apply multi-agent mirror descent to find the approximate quality profile required in Line 4 in $O(1/\delta)$ iterations [12], each requiring $O(n)$ time to evaluate creator gradients. The δ -FSE condition can also be efficiently verified. For each creator, it suffices to solve a single one-dimensional optimization: maximize utility over $x_i \geq 0$ with $s_i = 0$, holding the strategies of all other creators fixed. This follows from Lemma 2, which ensures that any profitable deviation is dominated by a deviation that has $s_i = 1$ or $s_i = 0$. The former is implicitly covered by the approximation guarantee of the computed profile $\mathbf{x}$, and the latter can be efficiently approximated, as we demonstrate that the utility function is Lipschitz continuous under each fixed value of s . We choose δ' small enough so that approximation errors from the ESE computation and the deviation check are jointly bounded by δ . After finding the maximal utility under $s_i = 0$, it remains only to test whether the resulting gain exceeds δ , thereby completing the verification.

5 Experiments

In this section, we conduct experiments to understand equilibrium outcomes and, more generally, how the allocation parameter ρ impacts creator behavior. We consider cost functions of the form $c_i(x) = a_i x^\theta$. In the default configuration, the parameters are $n = 20$, $\gamma = 0.8$, $\mu = 100$, $\theta = 1.5$, $\alpha = 0.5$, and the coefficients $\{a_i\}_{i=1}^n$ are independently drawn from the uniform distribution $\text{Uni}[1, 10]$. In addition to the default configuration, we varied one parameter at a time while keeping all others fixed at their default values to assess its individual impact. Each point in Figure 2 represents the average outcome across 150 independent game instances, with error bars indicating 95% confidence intervals obtained via bootstrapping. We discretized the possible values of ρ into 100 equally-spaced points in the range $[0, 1]$. For each ρ , we check if it induces an ε -FSE in $G(\rho)$ for $\varepsilon = 10^{-4}$. We ran all the executions on a standard

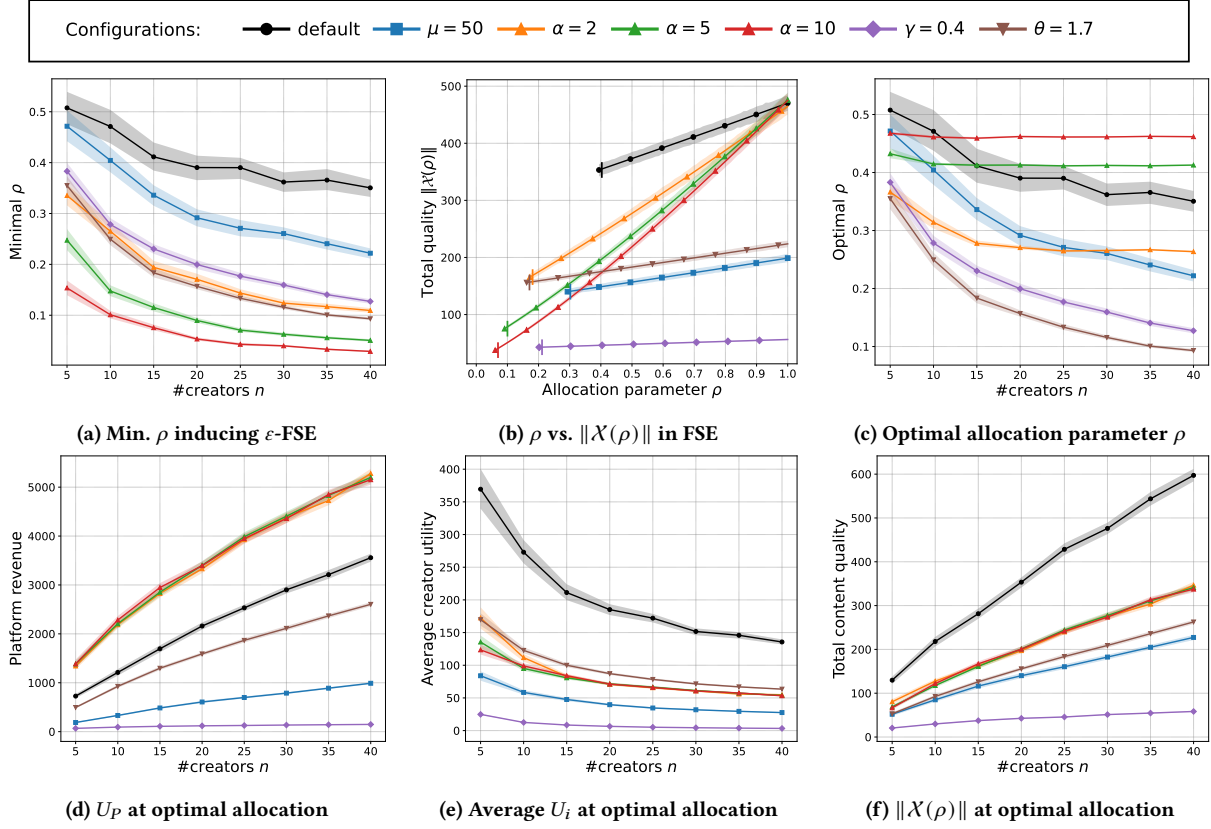


Figure 2: Revenue-allocation effects on equilibria, quality, revenue, and utilities.

laptop, completing the entire process in a few hours. We include all implementation details in Section A.

Minimal revenue-allocation threshold. Figure 2a illustrates how the minimal allocation parameter ρ required for inducing an FSE varies with ecosystem parameters. All curves decrease with the number of creators, as Theorem 5 implies: the content of each creator becomes insignificant w.r.t. all other created content, so withholding has a high alternative cost. Furthermore, more powerful GenAI (larger α) enables lower ρ choices for stable FSE, as the curves associated with a change in α (red, green, orange, and black curves) are ordered. The black curve ($\mu = 100$) lies above the blue curve ($\mu = 50$), suggesting that creators demand more compensation as traffic increases.

Content quality vs. allocation parameter. Figure 2b shows how different values of ρ affect $\|\chi(\rho)\|$, the total content quality in FSE. Since not all values of ρ induce an FSE (see Figure 2a), each line begins at the minimal value of ρ for which an FSE exists under the corresponding parameter setting. As expected, larger values of ρ encourage creators to produce more content regardless of the setting, as they get better compensation from the platform. Notably, more powerful GenAI (larger α) reduces creator content production. Furthermore, different values of α align as ρ approaches 1. To see why, recall that if the platform allocates most GenAI-driven revenue to creators, creators do not suffer from the competition with GenAI.

Optimal allocation. Figure 2c presents the platform’s optimal allocation parameter. We see that all the curves are non-increasing, implying that gradually smaller allocation parameters are enough. Focusing on the impact of high GenAI quality parameter α (red, green, and orange curves), we see an interesting pattern: the allocation parameter plateaus as we increase the number of creators. That is, the platform’s optimal choice tends toward higher ρ regardless of ecosystem size, with larger α corresponding to larger optimal ρ . We find this surprising as Figure 2a indicates that more powerful GenAI permits lower values of the allocation parameter to induce an FSE. This counterintuitive finding indicates that platforms with more powerful GenAI must compensate creators more generously in terms of the allocation. For weaker GenAI, the optimal allocation parameter is similar to the minimal allocation threshold required to induce an FSE: for $\alpha = 0.5$, the black curves in Figure 2a (minimal parameter) and Figure 2c (optimal parameter) are almost identical.

Platform-optimal equilibrium outcomes. Figures 2d–2f examine the platform’s revenue, creator utilities, and total quality under the platform-optimal allocation. As expected, larger creator populations increase platform revenue and total quality (increasing curves in Figure 2d and Figure 2f) while decreasing the average creator returns (decreasing curves in Figure 2e). Interestingly, powerful GenAI (red, green, and orange curves) yield nearly identical patterns in all graphs, suggesting that beyond a certain threshold,

more powerful GenAI does not significantly impact economic outcomes. Moreover, weak GenAI (black curve) is bad for the platform compared to powerful GenAI, and good for creator utility and total content quality.

6 Extensions

In this section, we provide three natural extensions of our framework and discuss their implications on our main results. We present additional extensions in Section B.

6.1 Multiple Topics

The following extension enables creators to allocate different efforts across various content topics, while the platform maintains a single revenue-allocation rule. Consider K distinct content topics. Each creator i now chooses a quality vector $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,K}) \in [0, \infty)^K$ and a sharing vector $\mathbf{s}_i = (s_{i,1}, \dots, s_{i,K}) \in [0, 1]^K$, where $x_{i,k}$ and $s_{i,k}$ represents their quality and sharing level in topic k , respectively. The cost function, which now gets $\mathbf{x}_i$ as input, is convex but need not be separable across topics (i.e., $c_i(\mathbf{x}_i) = \sum_k c_{i,k}(x_{i,k})$).

Each topic k generates independent user traffic T_k , and the GenAI system's quality is $Q_{AI,k}$; both depend only on the restriction of $\mathbf{x}$ and $\mathbf{s}$ to topic k and have the same functional forms as in the single-topic case. The platform commits to a single allocation parameter $\rho \in [0, 1]$ across all topics. Under proportional allocation, creator i 's revenue from topic k is given by

$$R_{i,k}(\mathbf{x}, \mathbf{s}; f_\rho) = \frac{T_k x_{i,k}}{\sum_{j=1}^n x_{j,k} + Q_{AI,k}} + \frac{T_k Q_{AI,k}}{\sum_{j=1}^n x_{j,k} + Q_{AI,k}} \cdot \frac{\rho x_{i,k} s_{i,k}}{\sum_{j=1}^n x_{j,k} s_{j,k}},$$

and its overall utility is $U_i(\mathbf{x}, \mathbf{s}; f_\rho) = \sum_{k=1}^K R_{i,k}(\mathbf{x}, \mathbf{s}; f_\rho) - c_i(\mathbf{x}_i)$.

Results. Lemma 2 extends to each topic independently: for fixed quality choices, the optimal sharing decision $s_{i,k}$ for topic k follows the same threshold structure in Equation (2), with $\tau_{i,k} = \frac{x_{i,k}}{x_{i,k} + \sum_{j \neq i} x_{j,k} + \alpha_k \sum_{j \neq i} x_{j,k} s_{j,k}}$. Lemma 3 holds as well—since the only interplay between different topics is through the cost function, the same analysis still applies. Theorem 5 requires some modifications, but the general idea remains the same: a large enough ρ admits FSE. The main technical difference is establishing per-topic rational quality bounds $\mathbf{x}_{i,k}^{\max}$ when the cost functions couple topics together. The approach of Algorithm 1 generalizes to this setting too, but testing the condition in Line 5 becomes computationally intensive, as we need to check each of the $2^K - 1$ possible sharing deviations. This can be avoided when the costs are separable across topics.

6.2 Prior GenAI Data

Next, we allow the GenAI system to possess a fixed body of prior data, independent of the creators' sharing contributions. Formally, we introduce a non-strategic creator indexed by 0 with fixed quality $x_0 > 0$ and full sharing level $s_0 = 1$. The quality of the GenAI system then becomes $Q_{AI}(\mathbf{x}, \mathbf{s}) = \alpha \left(x_0 + \sum_{j=1}^n x_j s_j \right)$, so that even if no strategic creator shares content, the GenAI still attains positive quality αx_0 . This modification also affects user traffic. Essentially, we assume that there is another source of x_0 body of content in the platform that is always available to the GenAI system.

Results. From creator i 's perspective, the new creator is just another content contributor with fixed quality and full sharing;

thus, the threshold from Lemma 2 updates accordingly to $\tau_i = \frac{x_i}{x_i + \|\mathbf{x}_{-i}\| + \alpha x_0 + (1+\alpha)x_0}$. Notably, prior data $x_0 > 0$ strictly decreases τ_i , making full sharing more attractive for creators. The proof of Lemma 3 requires a minor modification, as the dummy creator's strategy is fixed. Nonetheless, the resulting subgame among the creators maintains the DSC property. All other results extend directly to this setting.

6.3 Alternative Optimization Objectives

While our analysis in Section 4 focused on maximizing platform revenue (recall (PO)), there are other desirable objectives to consider. For instance, a platform may wish to balance its retained value with total content production and creator welfare. Our framework naturally extends to such cases, where the target function is Lipschitz continuous in the quality profile $\mathbf{x}$ and the allocation parameter ρ . The platform's problem becomes

$$\sup_{\rho \in [0,1]} f(\mathcal{X}(\rho), \rho) \quad \text{subject to } (\mathcal{X}(\rho), 1) \text{ is an FSE.}$$

Natural choices for f include total content production $f(\mathbf{x}, \rho) = \|\mathbf{x}\|_1$, creator welfare $f(\mathbf{x}, \rho) = \sum_{i=1}^n U_i(\mathbf{x}, 1; f_\rho)$, or regularized combinations such as $f(\mathbf{x}, \rho) = (1 - \lambda)U_P(\mathbf{x}, 1; f_\rho) + \lambda\|\mathbf{x}\|_1$.

Results. The same algorithmic approach from Algorithm 1 applies directly to these generalized objectives, with the approximation guarantees of Theorem 8 carrying over after adjusting the constants A and B to account for the Lipschitz properties of f .

7 Discussion and Future Work

We explored strategic data sharing and revenue allocation in content platforms with GenAI. We modeled the interaction between creators and the platform, where each creator chooses both the quality of their content and the extent to which they share it with the platform. The platform, which commits to an allocation rule, allocates them a portion of its GenAI-driven revenue based on their content qualities and sharing decisions. We characterized conditions for inducing an FSE, and proposed an efficient approximation algorithm for solving the platform's revenue maximization problem (PO). Our simulations examined sensitivity and equilibrium outcomes.

We see several promising directions for future research. First, while our analysis focused on proportional allocation rules, other functional forms may yield better outcomes. For instance, allocation rules that exhibit a staircase structure, providing high revenue for certain quality levels but diminishing afterward, could potentially yield more nuanced and efficient outcomes. Second, we restricted our attention to FSEs, but other PNEs (non-FSEs) may exist. Indeed, Lemma 2 hints that an asymmetric sharing, where only some creators choose to share content, can occur in equilibrium. Understanding which allocation rules can induce such equilibria and whether they lead to improved platform revenue remains an important open question.

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A Omitted Simulation Details

This section provides technical details about the algorithms and computational resources used in our experimental simulations.⁴

A.1 ESE Computation

Theoretical algorithm. Algorithm 2 provides a theoretically justified method for approximating the unique ESE for a given revenue-allocation parameter ρ . It employs multi-agent mirror descent with gradient-based updates to iteratively refine each creator's quality choice x_i while maintaining non-negativity constraints. The strong monotonicity of the enforced sharing game (established in Lemma 3) guarantees convergence at a rate of $O(1/T)$ [12], where T is the number of iterations.

Practical implementation. While Algorithm 2 provides provable convergence guarantees, we adopted a more practical approach in our simulations. Since each creator's utility is strictly concave in their own quality x_i (as established in Lemma 3), the ESE is uniquely characterized by the first-order conditions— for each creator $i \in [n]$, the equilibrium quality satisfies $\left. \partial U_i / \partial x_i \right|_{(x^*, 1)} = 0$. We solve this system numerically using `scipy.optimize`'s highly-optimized root-finding routines. Under our cost assumptions, interior solutions are

⁴Our code is available at https://github.com/GurKeinan/Code_for_Strategic_Content_Creation_with_GenAI_To_Share_or_Not_to_Share_Paper

Algorithm 2 Multi-Agent Mirror Descent for ESE Quality Profile (adapted from Bravo et al. [12])

Input: Number of creators n , cost functions $\{c_i\}_{i=1}^n$, model parameters $\mu, \alpha, \gamma, \rho$, number of iterations T , learning rate schedule $\{\eta_t\}_{t=0}^{T-1}$, initial quality profile $x^{(0)} \in \mathbb{R}_{\geq 0}^n$.

Output: Approximate ESE quality profile $x^{(T)}$.

- 1: Initialize quality profile $x \leftarrow x^{(0)}$.
 - 2: **for** $t = 0$ to $T - 1$ **do**
 - 3: Calculate gradient vector $v^{(t)} \in \mathbb{R}^n$. For each creator i :
 - 4: Let $X^{(t)} = \sum_{j=1}^n x_j^{(t)}$ and $X_{-i}^{(t)} = X^{(t)} - x_i^{(t)}$.
 - 5: $v_i^{(t)} \leftarrow \frac{\mu(1+\alpha\rho)}{1+\alpha} (X^{(t)})^{\gamma-2} [\gamma x_i^{(t)} + X_{-i}^{(t)}] - c'_i(x_i^{(t)})$.
 - 6: Update candidate profile: $x_{\text{cand}} \leftarrow x^{(t)} + \eta_t v^{(t)}$.
 - 7: Project onto feasible set: $x^{(t+1)} \leftarrow \max(0, x_{\text{cand}})$.
 - 8: **return** $x^{(T)}$.
-

Algorithm 3 FSE Stability Check via $s_i = 0$ Deviation

Input: Candidate ESE quality profile $x^* \in \mathbb{R}_{>0}^n$, cost functions $\{c_i(x_i) = a_i x_i^{\theta_i}\}_{i=1}^n$, model parameters $\mu, \alpha, \gamma, \rho$, stability tolerance $\varepsilon_{\text{tol}} > 0$.

Output: Boolean indicating if $(x^*, 1)$ is stable (True) or not (False).

- 1: **for** $i = 1$ to n **do**
 - 2: Calculate utility at candidate profile: $U_i^* \leftarrow U_i(x_i^*, 1; x_{-i}^*, 1_{-i})$.
 - 3: Calculate rational quality upper bound for deviation: $x_i^{\text{max}} \leftarrow (\mu/a_i)^{1/(\theta_i-\gamma)}$.
 - 4: $U_{\text{dev}}(s_i = 0) \leftarrow \max_{x'_i \in [0, x_i^{\text{max}}]} \{U_i(x'_i, 0; x_{-i}^*, 1_{-i})\}$. ▷
 - 5: Requires 1D solver
 - 5: Check stability condition:
 - 6: **if** $U_{\text{dev}}(s_i = 0) - U_i^* > \varepsilon_{\text{tol}}$ **then**
 - 7: **return** False ▷ Player i has a profitable deviation
 - 8: **return** True ▷ No player has a profitable deviation to $s_i = 0$
-

guaranteed, and the strict concavity ensures that any profile with near-zero gradients closely approximates the true ESE.

A.2 FSE Stability Check

Algorithm 3 verifies whether a given profile is FSE by checking if any creator can profitably deviate to not sharing ($s_i = 0$) (necessary and sufficient due to Lemma 2). For each creator, it solves a one-dimensional optimization problem to find their best response when withholding content. The profile is deemed stable only if no creator benefits from such deviation beyond the tolerance threshold ε_{tol} . This method is similar to the one we described for Algorithm 1, only we used a standard 1D solver rather than a grid search.

A.3 Simulation Pipeline and Resources

For each parameter combination we checked, we initialized 150 independent game instances with randomly sampled cost coefficients. For each instance, we evaluate 100 equally-spaced values of $\rho \in [0, 1]$. The computational workflow proceeds as follows: Algorithm 2 computes the unique ESE for each ρ value, leveraging the strong monotonicity properties established in our theoretical analysis. Subsequently, Algorithm 3 verifies whether each computed

ESE also constitutes an FSE by checking for profitable deviations to non-sharing strategies. Results are averaged across instances with 95% confidence intervals computed via bootstrapping.

The complete simulation suite was executed on a standard MacBook Pro with 18 GB of RAM and 8 CPU cores. We use parallel processing across 6 cores to accelerate computation. The full set of experiments required approximately 6 hours of computation time. The breakdown includes: ESE computation and FSE verification across all parameter combinations, which took about 6 hours in total; and statistical analysis and bootstrapping for confidence intervals, plotting, and visualization, which took less than 1 minute. The implementation uses standard Python scientific computing libraries (NumPy, SciPy) without GPU acceleration, as the algorithms are exclusively CPU-based.

B Further Extensions

B.1 GenAI-Dependent Traffic

In our main model, user traffic $T(\mathbf{x})$ depends solely on the quality of human-generated content. However, high-quality AI-generated content may also attract users to the platform. To capture this effect, we extend our traffic model to incorporate both human and AI contributions:

$$T(\mathbf{x}, \mathbf{s}) = \mu (\|\mathbf{x}\| + Q_{\text{AI}}(\mathbf{x}, \mathbf{s}))^\gamma.$$

All other components of the model remain unchanged.

Results. Under this extended traffic model, the creator's optimal sharing decision retains its threshold structure, with an adjusted threshold given by $\tau_i = \frac{1}{\alpha} \left[\left(\frac{\sum_j x_j + \alpha \sum_j x_j s_j}{\sum_j x_j + \alpha \sum_{j \neq i} x_j s_j} \right)^{1-\gamma} - 1 \right]$. Notably, larger values of γ decrease τ_i , making sharing more attractive. This occurs because higher traffic elasticity amplifies the positive externality that sharing exerts on the platform's traffic.

The analysis of ESE remains tractable under this extension. In the enforced sharing game where $\mathbf{s} = \mathbf{1}$, the traffic function simplifies to $T(\mathbf{x}, \mathbf{1}) = \mu(1 + \alpha)^\gamma \|\mathbf{x}\|^\gamma$. This expression is structurally identical to our original traffic model, differing only by a multiplicative scaling of the baseline parameter μ by $(1 + \alpha)^\gamma$. Consequently, Lemma 3 applies directly: the subgame $G(\rho)$ under enforced sharing admits a unique ESE. Since the utility function at $\mathbf{s} = \mathbf{1}$ retains its structural properties, our main results carry over to this setting with adjusted constants. Specifically, the FSE existence condition (Theorem 5) holds by redefining $x_i^{\max}$ as the unique positive solution to $c_i(x) = \mu(1 + \alpha)^\gamma x^\gamma$. The approximation guarantees of Theorem 8 follow similarly.

B.2 General GenAI Quality Function

In our baseline model, the quality of the GenAI system is linear in the aggregate amount of shared content, given by $Q_{\text{AI}}(\mathbf{x}, \mathbf{s}) = \alpha \sum_{i=1}^n x_i s_i$. This specification is both analytically convenient and captures the essence of the strategic dynamics in the system. However, in practice, GenAI systems often exhibit diminishing marginal returns as the amount of training data grows [30]. To capture this effect, we consider the following generalization of Q_{AI} :

$$Q_{\text{AI}}(\mathbf{x}, \mathbf{s}) = \alpha \left(\sum_{i=1}^n x_i s_i \right)^\beta, \quad \beta \in [0, 1].$$

This form, introduced by Yao et al. [48], captures diminishing returns in GenAI quality. Notice that when $\beta = 1$, this formulation reduces to our original model. Most of our results cannot be directly applied to this setting. However, we present the results that somewhat carry over. We refer the reader to the full version of this paper for simulations examining the impact of β , which indicate that the qualitative trends observed in the linear case persist under this generalization [31].

The following is a weaker version of Lemma 2. Although it does not provide a full characterization of the best sharing responses of the creators, it does provide the same Prisoner's Dilemma-like intuition discussed in the main text—profiles where no creator shares data are unstable even for not-so-large values of ρ .

PROPOSITION 11. *Let $(\mathbf{x}, \mathbf{0})$ be a strategy profile such that $x_i > 0$ for every $i \in [n]$. If $\rho > \frac{x_i}{\|\mathbf{x}\|}$ for at least one creator $i \in [n]$, then $(\mathbf{x}, \mathbf{0})$ is not a PNE.*

This result captures the Prisoner's Dilemma intuition: even modest revenue sharing can destabilize non-sharing equilibria, as individual creators face strong incentives to be the first to share and capture AI revenue.

Another result that can be extended is Proposition 4. Turns out that it can be directly extended to this setting without the need for additional assumptions or relaxations. The main technical difference is in proving that full sharing is still the best response for all creators, as Lemma 2 cannot be applied in this setting. After establishing that, it turns out that under full sharing, the players' utility simplifies to a function that does not depend on β anymore, and we can employ Lemma 3 to guarantee the existence of a PNE.

B.3 Other Allocation Rules

In this subsection, we analyze two alternative functional forms to the proportional allocation rule we studied throughout the paper (recall Equation (1)). We demonstrate that not only the sheer proportion of allocated revenue, but also how revenue is distributed among creators, significantly influences creators' behavior. Despite their apparent appeal, some allocation rules may not induce FSE even when the majority, if not all, of the GenAI-driven revenue is shared with creators. Consider the two following allocation rules parameterized by revenue share $\rho \in [0, 1]$:

- (1) *Winner-Takes-All (WTA):* The ρ -fraction of AI revenue is given to creators providing maximal effective contribution. Let $W(\mathbf{x}, \mathbf{s}) = \arg \max_{j \in [n]} \{x_j s_j\}$. Then:

$$f_{i,\rho}^{\text{WTA}}(\mathbf{x}, \mathbf{s}) = \begin{cases} \rho / |W(\mathbf{x}, \mathbf{s})| & \text{if } i \in W(\mathbf{x}, \mathbf{s}) \text{ and } \max_j x_j s_j > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (2) *Binary Threshold with Equal Shares (BTES):* The ρ -fraction is shared equally among creators who fully opt-in ($s_i = 1$):

$$f_{i,\rho}^{\text{BTES}}(\mathbf{x}, \mathbf{s}) = \begin{cases} \rho / |\{j: s_j = 1\}| & \text{if } s_i = 1 \\ 0 & \text{otherwise} \end{cases}$$

While our main results demonstrate that the proportional allocation rule admits FSEs for sufficiently high values of ρ (Proposition 4, Theorem 5), the following proposition shows that such stability guarantees do not extend to these alternative allocation rules.

PROPOSITION 12. *Under either WTA or BTES allocation rules with $\alpha > 0$ and $n \geq 2$, there exist instances where no FSE exists, even with full revenue redistribution ($\rho = 1$).*

The underlying issue is that neither rule properly aligns individual contributions with individual allocation, an issue that the proportional allocation rule solves implicitly.

C Modeling Assumptions and Limitations

This section outlines key assumptions in our modeling framework and discusses their implications and limitations.

User Traffic Modeling. Our baseline model assumes that platform user traffic is driven entirely by human-created content, with AI-generated content affecting allocation but not the traffic volume. This encompasses the scenario of a fixed number of users as a special case when $\gamma = 0$. In Section B, we relax this assumption by considering a traffic model that incorporates both human and AI-generated content, and show that our main results carry over with appropriately adjusted constants. More general traffic models—such as those capturing nonlinear interactions between human and AI quality—remain unexplored.

Generative AI Quality Modeling. We model the quality of GenAI content as a linear function of the total shared creator data. In practice, the relationship between training data quantity and model quality typically exhibits diminishing returns, consistent with established scaling laws in machine learning [30]. To address this, Section B considers a generalized quality function $Q_{AI} = \alpha(\sum_i x_i s_i)^\beta$ with $\beta \in [0, 1]$. While certain theoretical guarantees valid under the linear model do not hold in this generalized setting, our simulations suggest that the key qualitative insights persist. Other factors that may affect GenAI quality in practice—such as data diversity, representativeness, and complementarity among creators’ contributions—are not captured by our model and remain directions for future work.

Scalar Representation of Content Competitiveness. Each creator’s quality choice is represented as a single scalar $x_i \in [0, \infty)$, embodying the overall “competitiveness” of their content. This quantity encompasses various dimensions, including quality, quantity, trend alignment, expressiveness, domain specificity, and novelty. This modeling choice is a common simplification in the literature [48, 50], enabling tractable analysis while preserving the essence of competitive dynamics. However, real-world scenarios may involve more intricate representations, such as continuous vectors of features [26], requiring entirely different analytical tools.

Continuous Data Sharing Decisions. Our model enables creators to select their sharing level continuously, represented by $s_i \in [0, 1]$. In practice, platforms usually provide binary options. Nevertheless, all our results remain valid under binary sharing choices, as Lemma 2 ensures that only binary sharing decisions are rational.

Static Game Formulation. Our model captures a one-shot interaction between the platform and creators. In practice, creator-platform relationships are ongoing, and the quality of GenAI systems depends on data accumulated over time. Such dynamic settings introduce richer strategic considerations, including the timing of sharing decisions and the platform’s ability to update its allocation

rule. Modeling these interactions as repeated or sequential games constitutes a natural extension of our framework.

D Omitted Proof Sketches and Details

PROOF SKETCH OF LEMMA 3. We apply the classic Theorem 2 of Rosen [34], which requires three key components: (i) compact and convex strategy spaces, (ii) strictly concave utility functions, and (iii) diagonal strict concavity of the game.

Although we formally define creator strategies over $[0, \infty)$, we show that any rational quality choice—i.e., a choice that guarantees non-negative utility—must satisfy $x_i \leq x_i^{\max}$, where $x_i^{\max}$ is the unique positive root of $\mu x^\gamma = c_i(x)$. This is due to the strong convexity of the cost functions.

The second component follows from differentiating the utility under enforced sharing, which yields a negative second derivative.

For the third component, we show that for distinct strategy profiles $\mathbf{x}, \mathbf{x}'$ it holds that $\sum_{i=1}^n (v_i(\mathbf{x}') - v_i(\mathbf{x}))(x'_i - x_i) < 0$, where $v_i(\mathbf{x})$ is the gradient of creator i ’s utility. We decompose this sum into a cost component, $-\sum_{i=1}^n (c'_i(x'_i) - c'_i(x_i))(x'_i - x_i)$, and a revenue component. The cost component is negative due to the strict convexity of the cost functions. Demonstrating that the revenue component is non-positive necessitates a more thorough analysis. $\square$

Verification of Strong Monotonicity. To establish that the enforced sharing game is strongly monotone under strongly convex cost functions, it suffices to check that

$$\sum_{i=1}^n (v_i(\mathbf{x}') - v_i(\mathbf{x}))(x'_i - x_i) \leq -\sigma \|\mathbf{x}' - \mathbf{x}\|_2^2$$

for some $\sigma > 0$. As shown in the proof of Lemma 3, the revenue component of this expression is non-positive. Meanwhile, strong convexity of the costs implies

$$\sum_{i=1}^n (c'_i(x'_i) - c'_i(x_i))(x'_i - x_i) \geq \beta \|\mathbf{x}' - \mathbf{x}\|_2^2.$$

Thus the inequality holds with $\sigma = \beta$, completing the verification.

Note on Lipschitz Continuity. The proof of Theorem 8 heavily relies on the Lipschitz continuity of the platform’s revenue function and the players’ utilities w.r.t. both ρ and $\mathbf{x}$. A technical drawback is that these functions are not Lipschitz near $\mathbf{x} = 0$. To address this, we construct a positive lower bound on the total quality in any ESE, $\|\mathbf{x}(\rho)\|$, which ensures that the domain of these functions is bounded away from zero for all ρ . Together with an upper bound on $\|\mathbf{x}(\rho)\|$, we can apply the Mean Value Theorem to derive Lipschitz constants for the relevant functions.